

Quantum Computing

Part 9 - Industry, Academia & Claims: Academic & Theoretical

Compiled by the Spaced Research Ledger

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This part covers theory rather than machines. It records new algorithms and proofs about what quantum computers can and cannot do faster than classical ones. It also records the steady work of showing that some claimed quantum advantages were never real.

That last category deserves its own word. Dequantization is the practice of taking a quantum algorithm that was supposed to be exponentially faster and finding a classical algorithm that matches it. It is a normal and productive part of this field, and Section 3 collects the recent cases. A quantum speedup that survives dequantization attempts is worth considerably more than one that has not been tested.

One conceptual thread runs through Sections 1 and 2. The most reliable quantum speedups come from problems with mathematical structure the algorithm can exploit. For unstructured problems, the recent theoretical results tighten the limits on what any quantum machine could achieve.

Section 1 covers new algorithms, Section 2 complexity and hardness results, Section 3 dequantization, Section 4 verification and benchmarking theory, Section 5 error-mitigation theory, and Section 6 negative results.

1. New and Improved Quantum Algorithms

Recent algorithm work concentrates on linear algebra, differential equations and optimization rather than on the codebreaking applications that made the field famous. Several results trade theoretical advantage for circuits that could actually be run.

Recent Developments

A peer-reviewed algorithm solves the Sylvester equation, a matrix equation of the form $A X + X B = C$, by constructing a block encoding of the solution matrix [1]. Its query and gate complexities are almost linear in a condition number from the input matrices. They are logarithmic in dimension and inverse error. It can solve BQP-complete problems efficiently, which places it among the hardest problems a quantum computer is believed to handle.

Researchers at the Chinese Academy of Sciences published an algorithm for topological data analysis on directed graphs, introducing a path-homology approach with a universal encoding protocol [2]. The guaranteed speedup depends on how the data is accessed, with exponential gains possible under certain conditions. That conditional phrasing is the norm in this area and should be read carefully.

A preprint develops an algorithm for the dynamics of the nonlinear viscous Burgers equation, with extensions to general fluid equations [3]. It combines Carleman linearization with a permutation matrix representation and a linear combination of Hamiltonian simulations. It scales with the off-diagonal norm rather than the full

matrix norm. Nonlinear equations are difficult for quantum computers because quantum mechanics is linear, so any progress here is notable in itself.

Current State

Several earlier results remain the reference points.

Improved algorithms for reinforcement learning combine value iteration with quantum mean estimation and quantum maximum finding to achieve speedups over classical methods [4]. A bounded-error algorithm for modular factorials breaks the exponent-one-half barrier for the first time, through exact reconstruction of a Jacobi sum in compact algebraic form [5]. A space-efficient algorithm solves the elliptic curve discrete logarithm problem over prime fields using only $3n+6 \lceil \log n \rceil$ qubits, reducing the qubit count for that construction [6].

Decoded Quantum Interferometry is an algorithm for a class of optimization problems [7]. For one special case there is strong evidence of a superpolynomial speedup over the best known classical methods for approximate solutions. A quantum dynamic-programming algorithm for the travelling salesman problem achieves a feasible circuit implementation using very few qubits, trading theoretical advantage for practical implementability [8].

One report captures how contested this area is. An algorithm posted as a preprint was reported to run faster than all known classical algorithms on a wide class of optimization problems [9]. No classical algorithm had matched it as of that report. The same report notes that a separate claimed speedup published in Science met classical rebuttals within days of publication.

2. Complexity and Hardness Results

These are proofs about limits rather than methods. The most important recent one narrows where unstructured quantum speedups can exist at all.

Recent Developments

A preprint titled Quantum Speedups Require Structure or Depth shows that every t -query, d -round quantum algorithm can be simulated on most inputs with $t^{O(d^2)}$ classical queries [10]. The implication is direct. For unstructured problems, a superpolynomial quantum speedup requires more than constant circuit depth, and an exponential speedup requires polynomial depth. The result is not yet peer-reviewed.

A second recent item requires a warning rather than a summary. OpenAI reported in August 2026 that an internal model solved 10 longstanding open problems across mathematics, quantum complexity theory and theoretical computer science [11]. The claimed results include quantum parallel repetition and hardness bounds for the closest vector problem. No detailed proofs were released and no peer-reviewed validation was announced. This is a company claim about unverified results and must not be read as settled mathematics.

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Four researchers resolved a two-decade-old open problem by identifying a computational problem that provably requires a quantum proof of correctness, with no classical proof sufficing [12]. The work received the best-paper award at the STOC 2026 conference.

A 2025 paper on one-way state generators proved that their existence follows from almost all computational pseudorandom quantum state generators [13]. It also showed that a certain class of these generators can be broken by quantum polynomial-time algorithms given access to a particular oracle. Work of this kind maps which quantum cryptographic building blocks depend on which others.

A 2025 result reproves the hardness of computing distances of quantum error-correcting codes, using simpler arguments based on hypergraph product codes [14]. It gives a direct reduction from the classical minimum-distance problem. The codes in question are the subject of Part 6.

3. Dequantization and Classical Simulation

Every result here removes a claimed quantum advantage. Collectively they are the strongest available check on the optimism in the rest of this collection.

Current State

A peer-reviewed 2026 paper dequantizes the algorithm for simulating short-time dynamics of classical linear systems with geometrically local interactions, showing that this specific case yields no exponential advantage [15]. The contrast is instructive, because prior work had established an exponential speedup for classical lattice dynamics with long-range interactions. The advantage lived in the long-range structure, not in the problem generally.

A dequantization analysis using random Fourier features shows that classical algorithms can practically dequantize existing quantum-kernel machine learning methods [16]. That widens the set of machine learning tasks for which no quantum advantage can be claimed.

A peer-reviewed paper presents a quantum-enhanced classical method for approximating regions of the expectation-value surface produced by parametrized quantum circuits [17]. A lower bound has separately been established on the classical simulation cost of star-network correlations, bounding how efficiently classical resources can reproduce that class of quantum correlations [18].

One item is contested rather than established. A 2026 preprint argues that Google's December 2024 random-circuit-sampling claim occupied a regime whose classical simulability was closed within ten months by a general theorem on noisy random circuits [19]. It also states the underlying samples have not been replicated end to end. This is a disputed interpretation, not a settled finding.

4. Verification and Benchmarking Theory

This section asks how anyone knows a quantum result is correct. Both entries measure the tools rather than the machines.

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Two 2026 preprint benchmarks cover 36 and 40 machine-checkable theorem-completion tasks in the Lean 4 proof system [20]. They test whether AI agents can construct formally verified proofs in quantum algorithms and quantum information theory. The best-performing model scored 60.4 out of 100 on one and 59.6 out of 100 on the other.

A 2025 preprint develops scalable proxy circuits, built from a restricted set of operations, that estimate application-level performance of a quantum computer [21]. It gives theoretical grounding for when such restricted benchmarks are enough to predict performance on the harder circuits that real applications need.

5. Noise and Error Mitigation Theory

Error mitigation is the set of techniques for extracting useful answers from noisy machines without full error correction. The theory in this section is mostly about its limits, and the limits are severe.

Recent Developments

A peer-reviewed paper introduces surrogate-enabled zero-noise extrapolation, which uses classical learning surrogates to perform the technique with only constant measurement overhead for families of parametrized circuits [22]. It reports accuracy comparable to the conventional method and was validated on tasks up to 100 qubits. Constant rather than growing overhead is the important claim, for reasons the Current State below makes clear.

Two further preprints appeared in August 2026, neither peer-reviewed [23]. One frames error suppression, mitigation and full correction as complementary layers of a single strategy for the period between today's noisy machines and early fault tolerance. The other proposes an AI-assisted mitigation framework that models decoherence as sequences of weak measurements.

Current State

The central limitation is a cost that grows exponentially. Standard techniques including probabilistic error cancellation, zero-noise extrapolation and Clifford data regression can substantially reduce estimation error [24]. All of them incur a sampling overhead that scales exponentially as circuit size or noise increases. That single fact motivates most of the other work in this section.

A peer-reviewed 2025 paper reduces that overhead for probabilistic error cancellation, through Pauli error propagation combined with classical preprocessing [24]. It cuts the cost substantially for a restricted class of circuits.

Theoretical work presented at a 2025 workshop established rigorous worst-case bounds showing that error mitigation is fundamentally limited in its scalability as circuit size and noise grow [25]. The same workshop's report describes the field as still formative after nearly a decade, largely heuristic and problem-specific, with very few practical guidelines for when to apply which technique [25].

Two methodological cautions round this out. Apparent improvements in zero-noise extrapolation can arise from post-processing arithmetic artefacts rather than genuine noise suppression, so benchmarking must distinguish the two [26]. Verifiable blind probabilistic error cancellation offers a method for reliable mitigation without requiring full trust in the noise model or in the hardware performing the computation [27].

6. Negative Results

Published negative results are rarer than they should be in any field, and they are the most useful entries in this part.

Current State

No quantum advantage was found for fidelity-kernel methods in predicting the cross-section of stock returns, evaluated across 170 walk-forward windows on a point-in-time universe [28]. Finance is among the most heavily promoted quantum application areas, which makes a clean negative result there worth recording.

A claimed quantum speedup published in Science was publicly disputed within a week, with two separate research groups independently showing how to perform similar calculations classically [9].

A 2026 preprint contends that Google's Willow random-circuit-sampling result has not been replicated end to end [19]. It argues the demonstration occupied a region of circuit space later shown to admit classical simulation. The preprint frames this as a recurring pattern in which demonstrations were confined to classically simulable regimes by hardware constraints. That interpretation is contested and not universally accepted.

About This Report

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